



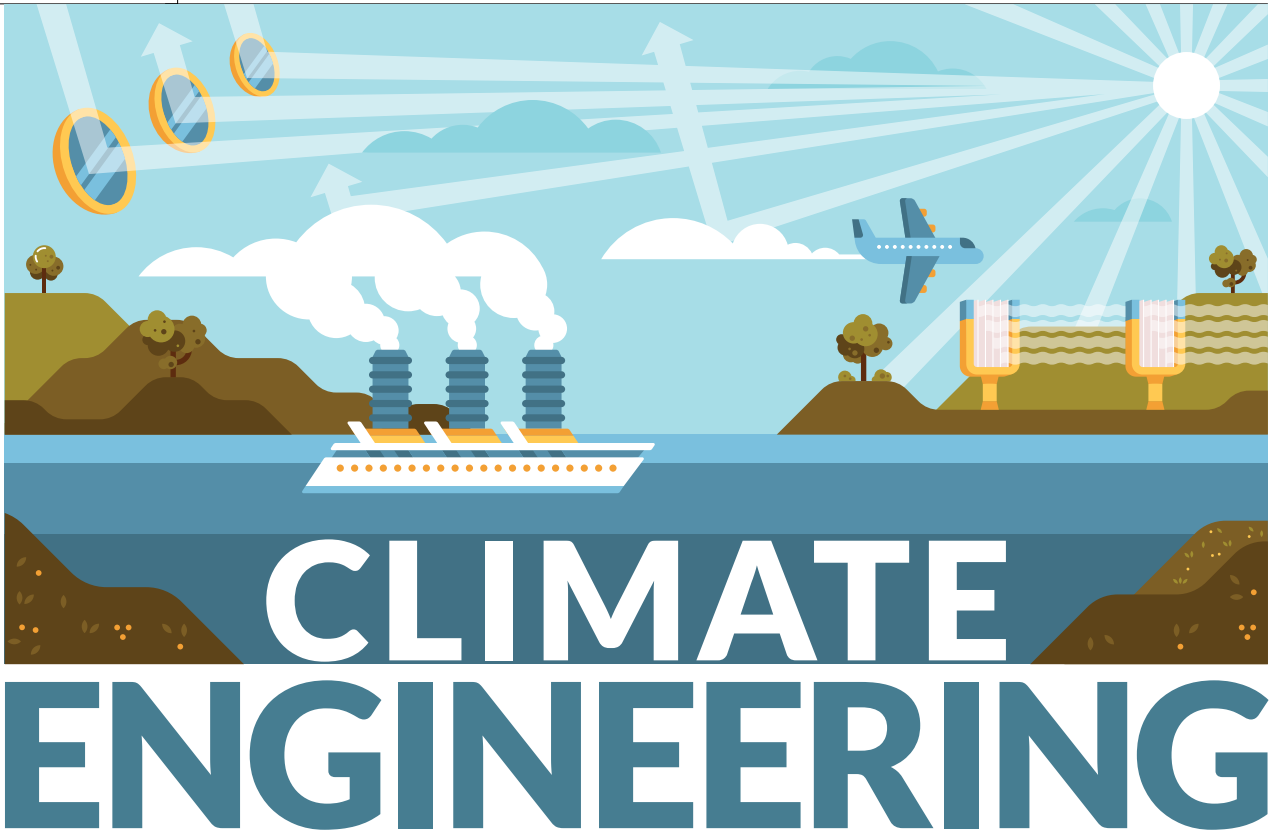
Arctic Warming & Earthquakes

Study suggests that the Arctic's rapid warming could have been triggered by a series of great earthquakes. <http://digit.in/jan21-01>



Robot Fleet

Australian researchers deploy deep-diving ocean robots to study how marine life captures and stores carbon. <http://digit.in/jan21-02>



What are the risks and rewards of technologies designed to reverse global warming?

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Climate engineering, commonly known as geoengineering is an umbrella term for all the techniques that can be used to reverse the global rise in temperatures, caused by industrial human activity. There are mainly two types of technologies, carbon dioxide removal, and solar radiation management (SRM).

The carbon dioxide removal technologies use fans and filters to mine the carbon dioxide out of the air, or dumping millions of tons of materials into the planet to initiate the processes that reduce the carbon

dioxide we have already dumped into the atmosphere. These range from iron pellets that can be dumped into the oceans to simulate plankton growth, that in turn can consume the carbon dioxide, or coating the planet with silicate minerals that can fix the carbon dioxide in the atmosphere.

The solar radiation management techniques sound even more ridiculous and outlandish. The idea is to reflect off as much energy as possible from the Sun, to prevent the heat from being trapped within the atmosphere. These include whitening mountaintops, brightening oceans with microbubbles, using genetic modification to increase the reflectivity of crops, pumping Sun blocking aerosols into the atmosphere, clearing forests over snowy terrain, and making clouds whiter.

One of the techniques suggests using drones to make the clouds in the upper atmosphere thinner, allowing more heat from the planet to escape into space, is considered to be an Earth radiation management technology instead of the solar radia-

tion management technology. All the climate engineering technologies are very much in the research stage, and most of the research only involve simulations to show how much heat or carbon dioxide can be reduced by the approach. The scientists are fully aware that many of these ideas sound impractical, but still scientists continue to research these technologies. The alternative to climate engineering is rapid energy transition, where the world moves on to better and cleaner alternatives to fossil fuels.

Rapid energy transition requires some major shifts in how the planet uses fossil fuels. Many of these shifts are underway, and it is just a matter of how slowly or quickly they take place. The main move is to convert most or all land transport into electric vehicles, instead of vehicles based on fossil fuels. This shift will be more difficult for goods ships and aeroplanes, but at least the fuel used for these can be made more efficient. Then there is a projected reduction in costs for energy storing materials, that is expected to give a boost to cleaner power genera-